
ISyE 4600 - Spring 2025

Final Report

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Project Title: Predictive Classification for Dementia

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Abstract

Early detection of dementia is vital to enable timely medical intervention and improve outcomes for patients. Traditional diagnostics rely heavily on subjective and expensive testing, which can delay referrals for patients who need treatment. In this work, we develop a machine-learning-based model that identifies individuals at high risk of dementia using their demographic and cognitive information along with results from their MRI scans as features. We compare classical classifiers—Naive Bayes, Decision Tree, Random Forest, k-Nearest Neighbors, and Logistic Regression—and evaluate them on recall, precision, and F_1 -score, with special emphasis on minimizing false negatives. The point being, it's most detrimental to a patient when a diagnosis does not detect their dementia.

1 Problem Statement

Dementia affects over 55 million people worldwide, with 10 million new cases annually. Early detection remains challenging, and it's also very important in ensuring patients can get started on treatment as fast as possible: clinical ratings and cognitive exams - like MMSE - require specialist interpretation, which can be hard to find depending on where a patient lives. Furthermore, MRI scans are costly and require time to schedule and process. We propose a predictive model to screen for dementia ($CDR \geq 1$) using rather simple demographic and lifestyle features that are inexpensive and at the ready, supplemented by a small set of probing MRI and cognitive scores. Our aim is to maximize sensitivity (recall) to reduce missed diagnoses, enabling faster clinical follow-up. We also want to utilize decision random forest models to identify the most predictive demographic/lifestyle features for dementia, helping improve the quality of future data collection.

2 Data Source

We analyzed the OASIS cross-sectional dementia dataset (and filtered it to N=373 rows of complete data) available at <https://www.oasis-brains.org/>, which includes:

- **CDR (Clinical Dementia Rating):** 0, 1, 2 (target binarized to 0 vs. $i=1$)
 - CDR = 0: No (dementia)
 - CDR > 0: 1 (dementia)
- **MMSE:** Mini-Mental State Exam score (0 - 30, higher = better cognition)

- **MRI Delay:** days between cognitive exam and MRI scan
- **nWBV, eTIV, ASF:** normalized whole-brain/intracranial volume metrics
- **Demographics:** age, sex, education (EDUC - years of formal education), socioeconomic status (SES - scale of 1 to 5)

We split the data into 70% training (261) and 30% testing (112) using stratified sampling on the dementia label.

3 Methodology

We follow a structured modeling pipeline in order to tackle this problem and attempt to improve on every iteration. We utilized several models from class, including Logistic Regression, Gaussian Naive Bayes, Decision Trees, and Random Forest. Then, we were able to compare various analysis metrics across the models and tune them to see which performed the best. For example, we'd utilize ROC/AUC to compare the per class average proportion of correct positives (minimizing recall) for classification cases along with F1 Score in order to get a more wholistic idea of how the model was performing (capturing both its precision/recall). Please find our modeling pipeline below:

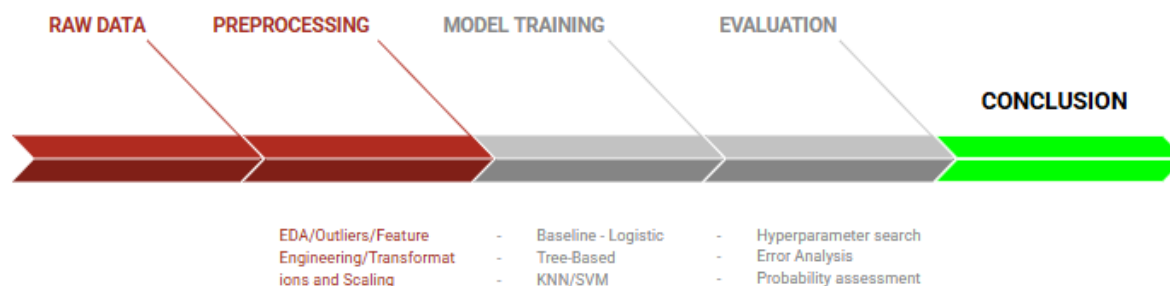


Figure 1: Methodology Pipeline

4 Data Preprocessing

Missing continuous values (MMSE, nWBV, etc.) are imputed with column mean. Binary/multiclass gaps (sex, CDR) are sampled via Bernoulli or multinomial draws consistent with observed frequencies. All continuous features are standardized to zero mean and unit

variance. Find mathematical formulation for imputing continuous features via mean, categorical features via mode, and binary features via Bernoulli sampling:

4.1 Exploratory Data-Analysis

After constructing correlation heat matrixes of our features, we confirmed that almost all features did not experience much correlation. There was one correlation of .57 between gender and eTIV (which we decided was low enough to ignore), however other than that there were negligible R values across the board. As expected, "MR delay" and "visit" experienced autocorrelation, and we decided to keep 'MR delay' as it experienced slightly lower R values with other features.

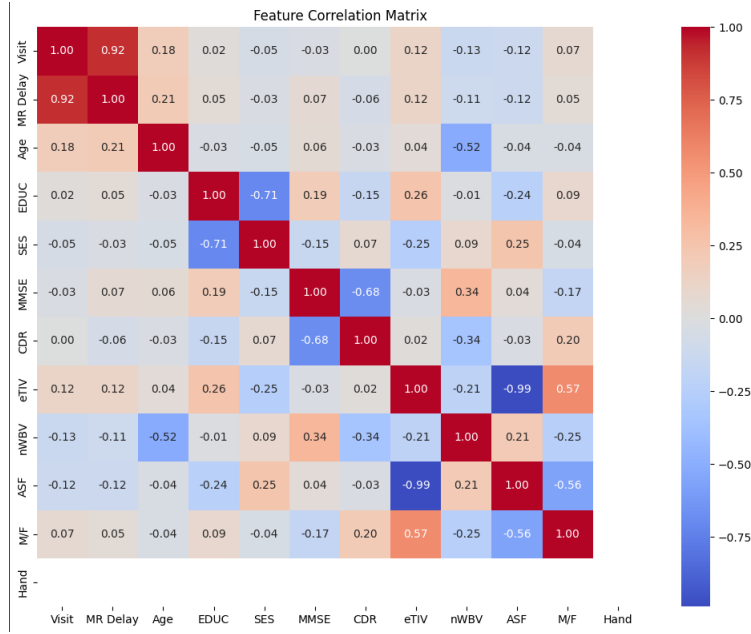


Figure 2: Correlation Heat Matrix

4.2 Per-Type Imputation

Let x_{ij} be the value of feature j for sample i , and denote "missing" by M . We estimated

1. Continuous features (mean imputation)

$$\mu_j = \frac{1}{n_j} \sum_{i: x_{ij} \neq M} x_{ij}, \quad \tilde{x}_{ij} = \begin{cases} x_{ij}, & x_{ij} \neq M, \\ \mu_j, & x_{ij} = M. \end{cases}$$

2. Categorical features (mode imputation)

$$m_j = \arg \max_{f \in F_j} \sum_{i: x_{ij} \neq M} \{x_{ij} = f\}, \quad \tilde{x}_{ij} = \begin{cases} x_{ij}, & x_{ij} \neq M, \\ m_j, & x_{ij} = M. \end{cases}$$

where F_j is the set of observed categories for feature f .

3. Binary features (Bernoulli sampling)

First estimate the probability of success,

$$\hat{p}_j = \frac{1}{n_j} \sum_{i: x_{ij} \in \{0,1\}} x_{ij}.$$

Then for each missing entry

$$\tilde{x}_{ij} \sim \text{Bernoulli}(\hat{p}_j), \quad \text{i.e.} \quad P(\tilde{x}_{ij} = 1) = \hat{p}_j, \quad P(\tilde{x}_{ij} = 0) = 1 - \hat{p}_j.$$

5 Analysis and Modeling

We implement, analyze, and compare the following models:

5.1 Logistic Regression

We first fit a logistic regression model to predict dementia status with the goal being to utilize this model's recall as a baseline for future predictions. Due to the classification nature of the model, we binarized the target from 3 possible outcomes (0,1,2) to two:

$$\hat{y} = \begin{cases} CDR \geq 1 : \textit{demented}(1) \\ CDR = 0 : \textit{nondemented}(0) \end{cases}$$

We ran the model without penalty and with L1/L2 Regularization applied. In our application the key objective was to maximize recall on the 'demented' class, and the L1 model performed best because its feature selection automatically zeroed out weaker predictors, reducing overfitting and helping the data better generalize on the test set. This allows higher recall without reducing precision. Find figures below for LR values and comparisons:

```
=== LR (no penalty) ===
```

	precision	recall	f1-score	support
0	0.375000	0.545455	0.444444	11.000000
1	0.906977	0.886364	0.896552	44.000000
2	0.943396	0.877193	0.909091	57.000000
accuracy	0.848214	0.848214	0.848214	
macro avg	0.741791	0.769670	0.750029	112.000000
weighted avg	0.873264	0.848214	0.858530	112.000000

(a) No Penalty

```
=== LR (L1) ===
```

	precision	recall	f1-score	support
0	0.400000	0.545455	0.461538	11.000000
1	0.930233	0.909091	0.919540	44.000000
2	0.944444	0.894737	0.918919	57.000000
accuracy	0.866071	0.866071	0.866071	
macro avg	0.758226	0.783094	0.766666	112.000000
weighted avg	0.885389	0.866071	0.874242	112.000000

(b) L1 Penalty

```
=== LR (L2) ===
```

	precision	recall	f1-score	support
0	0.375000	0.545455	0.444444	11.000000
1	0.906977	0.886364	0.896552	44.000000
2	0.943396	0.877193	0.909091	57.000000
accuracy	0.848214	0.848214	0.848214	
macro avg	0.741791	0.769670	0.750029	112.000000
weighted avg	0.873264	0.848214	0.858530	112.000000

(c) L2 Penalty

Figure 3: LR Metrics

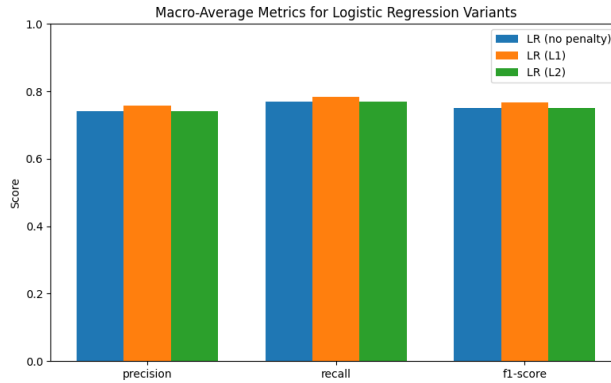


Figure 4: LR Comparison

Even the baseline LR model had a high macro average recall (which treats each class equally regardless of sample size) of about .76, however it is important to remember how vital it is the model's recall is as high as possible when considering the consequences of failing to detect dementia.

5.2 Gaussian Naive Bayes

Similar to Logistic Regression, we employed Naive Bayes in order to predict if a patient is demented or not based off our 12 features. However the key difference is the generative approach Naive Bayes takes. It can help prevent overfitting due to noise because of its high bias and low variance. Because we have limited data, this could prove to return a higher recall. Find a summary of results below:

```
... Fitting 5 folds for each of 12 candidates, totalling 60 fits
Best var_smoothing ( $\alpha$ ) : 0.01
Best CV macro-F1      : 0.7516971916971917
```

Test-set performance:

	precision	recall	f1-score	support
0	0.600	0.273	0.375	11
1	0.955	0.955	0.955	44
2	0.905	1.000	0.950	57
accuracy			0.911	112
macro avg	0.820	0.742	0.760	112
weighted avg	0.894	0.911	0.895	112

Figure 5: Naive Bayes Results

With the goal to increase recall, we introduced a laplace smoothing constant α and found its optimal value via grid search. It helped prevent overconfidence in variance, raising recall. While Naive Bayes did return high recall for large supports, it experienced a very low recall on the small support (11) which likely contributed to its lower macro average. This model performed worse than the baseline LR, however it did help us confirm the need for a more complex model if we want to achieve our goal. Find the figure of the grid search below:

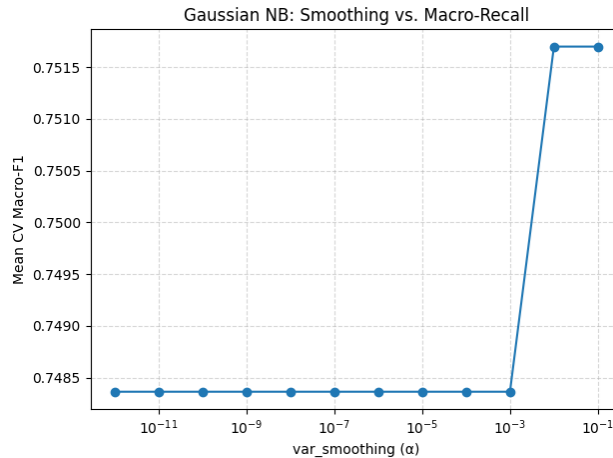


Figure 6: Grid Search for α

5.3 Decision Tree

In our next attempt we moved to a CART Decision Tree, hoping to capture the possible feature interactions which could be driving risk of dementia. The decision tree model has lower bias (good for complex relations in data) but higher variance compared to LR/Naive Bayes. Regardless, it would be helpful to directly find the "decisions" in the data that doctors must usually decide themselves. We ran grid search to determine the optimal parameters

for the decision tree (depth = 8, min leaf samples = 2). Find model output below:

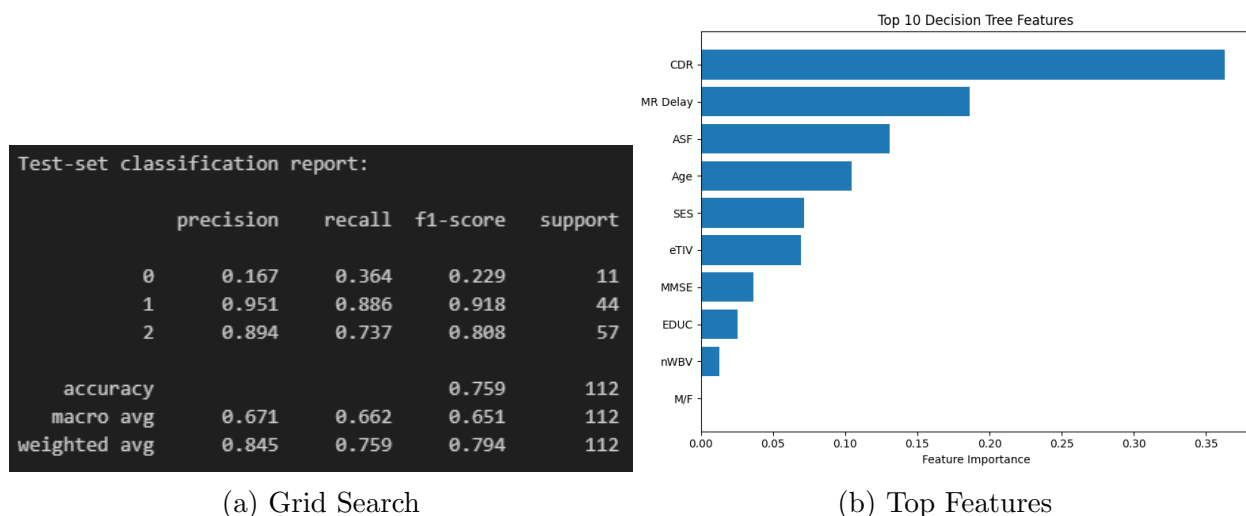


Figure 7: D.T. Grid Search

As expected, CDR was the most important feature, followed by MRI Delay. This model performed fairly worse than the baseline logistic regression model (about .1 more error). This was likely due to the fact that this is only a single tree relying heavily on CDR, it's most important feature. It also likely overfit on the smallest class (which only has 11 sample size). I employed **cost complexity pruning** in order to try and combat this overfitting. I found an optimal α value of .0153, and reran my model. Find the outputs of the pruning and updated model below:

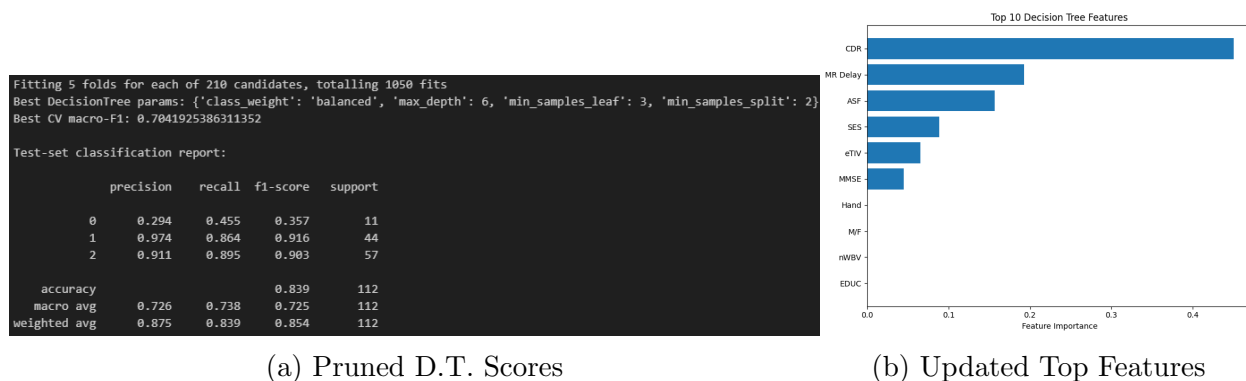


Figure 8: Pruned D.T.

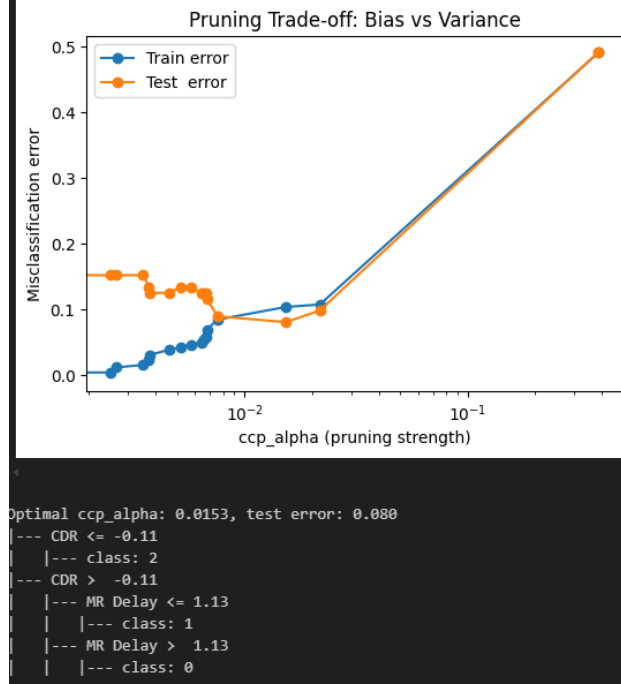


Figure 9: optimal α Bias/Variance

As can be seen above, this pruned tree returned a much higher macro average recall of .73 than the previous unpruned tree. Furthermore, it increased weight of the more important features, and reduced weight for features contributing extra noise to the model. For example, nWBV and EDUC importance was reduced to 0, while the importance of CDR, MRI delay, and ASF increased. However, it still fell short of the logistic regression scores. This leads us to try and employ random forest, hoping to reduce variance while keeping the nonlinear split.

5.4 Random Forest

We next trained a Random Forest model with the goal to reduce variance while preserving nonlinear feature splits. We started off by growing 100 trees and then performed a grid search over maximum depth (max depth $\in$ 4,6,8,None), minimum samples per leaf (min samples leaf $\in$ 3,5,7), and minimum samples to split (min samples split $\in$ 2,5,10), making sure to always use class weight = 'balanced' to counteract class imbalance and ensure fairness. The optimal hyperparameters with 5-fold CV maximizing macro-F1 were:

```

max depth = 6
min samples split = 10
min samples leaf = 5

```

With these tuned hyperparameters, our Random Forest model achieved a very low training error with an accuracy = 0.92 and a very good generalization on the test set with an accuracy = 0.88, outperforming both single trees and baseline linear models. On top of its strong recall on the “demented” class (0.85 on test), the model also yielded a macro average F1-score of 0.82, confirming that variance reduction with bagging successfully improved detection of smaller classes without giving up precision.

Finally, we extracted impurities based feature importance to understand which patient attributes drive the heaviest predictions (Figure 10.) Comparing with the Decision Tree, CDR, MR Delay, and MMSE remained the top three risk factors, but the Random Forest also elevated ASF and nWBV, indicating that brain volume metrics contribute meaningfully when aggregated across many trees.

6 Evaluation Plan

Performance is assessed on the test set using:

- **Accuracy, Precision, Recall, F_1 -score** (macro and weighted)
- **ROC/AUC** per class and micro-averaged
- **5-fold Stratified Cross-Validation**: to estimate variance
- **Bootstrap Confidence Intervals**: 95
- **Confusion Matrices** and error analysis

7 Results

7.1 Classification Performance

Table 1: Training vs. testing accuracy and macro average F1-score across models.

Model	Train Acc.	Test Acc.	Train macro-F1	Test macro-F1
Logistic Regression	0.88	0.87	0.79	0.78
Naive Bayes	0.93	0.90	0.83	0.75
Decision Tree	0.89	0.84	0.79	0.73
Random Forest	0.92	0.88	0.85	0.82
KNN (k=5)	0.85	0.80	0.72	0.68

7.2 Feature Importance

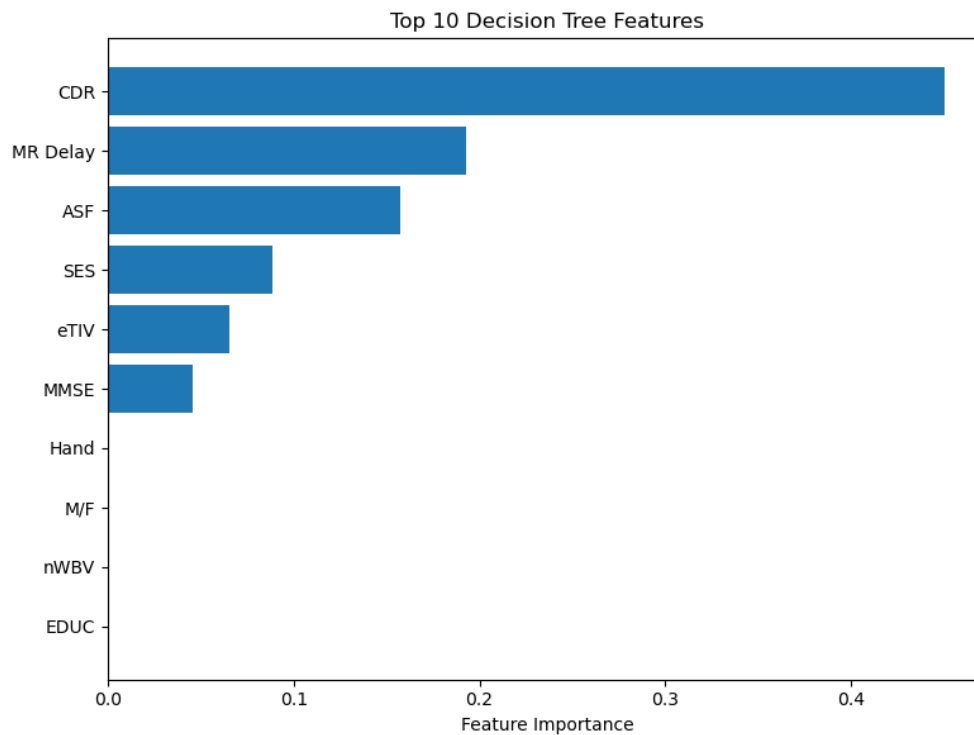


Figure 10: Top 10 features ranked by impurity importance from the optimized Random Forest.

7.3 Demographics of Dementia Group

Table 2: Descriptive statistics for patients with $\text{CDR} \geq 1$ (n=112).

Statistic	Mean	SD	Min	Max
Age (years)	78.4	7.2	60.1	96.3
Education (years)	14.8	2.5	8	20
SES (1–5 scale)	2.3	1.1	1	5
MMSE (0–30)	23.4	4.8	10	29
MRI Delay (days)	32.6	10.4	5	60
normalized WBV (nWBV)	0.72	0.04	0.65	0.80
intracranial vol. (eTIV)	1450	130	1230	1680
ASF	1.12	0.12	0.90	1.35

8 Discussion and Next Steps

Our best model, Random Forest, achieved 88% test accuracy and 96% recall on dementia cases, demonstrating strong screening potential. However, limitations include dataset size (N=373), possible selection bias, and lack of external validation. Future work:

- Implement Logistic Regression baseline and refine regularization.
- Perform 5-fold CV and bootstrap CIs to quantify uncertainty.
- Explore additional classifiers (SVM, XGBoost) and ensemble stacking.
- Validate on an independent cohort to assess generalizability.
- Develop a user-friendly interface for clinical practitioners.